INCH-POUND

MIL-PRF-32031
8 July 1998

PERFORMANCE SPECIFICATION
GRIT-BLAST MACHINE FOR COMPRESSOR HOUSING ASSEMBLY (T56 Series III)

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electrically and air-operated unit supported on a steel angle and channel iron frame and designed to provide a means for the removal of Nickel Aluminide Metco No. 406 or Stainless Steel Metcal No. 2 metal from coatings from the inside surfaces of the Compressor Case Housing.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be sent to: (SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Bldg 171, Kelly AFB, TX 78241-6524), by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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STANDARDS

FEDERAL

FED-STD-595 - Colors used in Government Procurement

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

GMC. Corp -Allison Div.

DWG 6796666 - Adapter-Compressor Housing Assy. Grit Blast
DWG 6796667 - Fixture-Holding Compressor Housing Assy. Grit Blast

(Application for copies should be addressed to the Engineering Data Repository, SA-ALC/TILDB, 485 Quentin Roosevelt Rd, Bldg 171, Kelly AFB, TX 78241-6524.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The grit blast machine shall be electrically and air-operated, supported on a steel angle and channel iron frame, suitably braced to carry the equipment that is to be mounted upon it. The compressor housing assembly shall be mounted to a rotation spindle permanently mounted inside the cabinet and shall be capable of being utilized with Allison's adapter, drawing number 6796666 or similar and Allison's holding fixture, drawing number 6796667 or similar. The compressor housing assembly shall revolve at a set speed while the grit blast nozzle cycles up and down on an elevator-type guide. The grit media of Metcolite C aluminum oxide No. 24 (20 to 50 mesh size) shall impinge on the inside surfaces at 50-80 PSI until the stainless steel bond is removed and a uniform appearance is obtained. The electrical control cabinet shall contain motor starters, fuses and relays, provided in accordance with standard NEMA enclosure that shall be mounted on the right-hand side of the grit blast cabinet.

3.2 Performance. The grit blast machine shall be provided with a turntable assembly that is capable of rotating the part, a ballscrew driven gun mover for the grit nozzle and a grit hopper assembly. A cartridge style dust collector shall be provided with an exhaust fan. The dust that is circulated in the

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cabinet from the grit blasting process is drawn from the inside of the cabinet through vents that are located near the top of the rear bulkhead of the cabinet by an external exhaust fan. Collected and purified air shall be exhausted through a vent to the atmosphere. The air exhausted from the dust collector must meet environmental regulatory standards of the South Coast Air Quality Management District (SCAQMD) with respect to particulate matter. The cartridges shall be capable of being cleaned by reverse pulsed compressed air. The cartridges shall be standard, commercially available products. Dust shall be capable of being collected in a barrel.

3.3 First article. When specified (see 6.2), one or more complete grit blast machine shall be subjected to first article inspection in accordance with 4.1.

3.4 Physical characteristics. The grit blast machine shall be a maximum of 84 inches in length, 80 inches in width, 120 inches in height.

3.5 Paint requirements. Unless otherwise specified (see 6.2), the grit blast machine shall be primer and painted to industries accepted standard practice for industrial machinery and FED-STD-595 colors used in government procurement.

3.6. Workmanship. The use of good commercial welding practices throughout shall be utilized. All welds shall be free of all slag and splatter. The unit's components shall be deburred and all sharp edges shall be broken.

3.7 Construction, fabrication and assembly. The interior of the chamber shall be lined with rubber and it shall extend into the door opening all the way around to ensure a good seal during the grid blasting cycle. The interior of the chamber walls shall be lined from the screen frame to the ceiling with commercial grade rubber.

3.8 Reliability. The service life (out-of-container) shall be up to 15 years with regular maintenance as specified by the manual.

3.9 Maintainability. The unit shall be capable of being maintained and operated with a minimum amount of hand tools and with commonly sized wrenches and tools.

3.10 Environmental requirements. The unit shall be designed to prevent degradation in order to prevent physical damage, such as corrosion, rust, loosening of finish, structural degradation, or deterioration of parts and materials under environmental conditions. The unit shall not be affected or damaged (i.e. corrosion and rust) by ambient temperature ranges of -60 F to 160 F, relative humidity of 5 to 99%, salt fog (for periods up to 48 hours) and dust (particles of 4 microns to 4000 microns).

3.11 Electrical requirements. The grit blast machine shall be wired to conform to NFPA 79. The machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed to operate on 3-phase, 60-hertz (Hz), dual 230/460-voltage and 15-20 KVA outlet. The machine shall be initially wired as specified (see 6.2).

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3.12 Safety. The grit blast machine design shall comply with OSHA Std 29CFR 1910.94 and ANSI Z9.4 for blasting operations and be such as to exclude the accumulations of dust and eliminate the potential for fires or explosions.

3.13 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 6.2), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Manufacturer's model designation.
d. Power input (volts, total amperes, phase, frequency).
e. Contract number or order number.
f. National stock number.
g. Date of manufacture.

3.14 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. The plate shall include the following information:

a. Points of lubrication.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. Verification.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified (see 6.2), the supplier shall furnish one or more grit blast machine's for first article inspection in accordance with 3.3. Unless otherwise specified (see 6.2) the first article inspection shall be an initial production item, which conforms to the requirements of this specification. Unless otherwise specified, first article inspection shall consist of the examination in 4.2.3 and all test methods and tests under section 4.2.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.1 Test sample. The first article inspection samples shall be representative of the production units the contractor intends to furnish.

4.2.2 Test report. Upon completion of the first article inspection, a test report in accordance with contractor's format, shall be provided to the government. The test report shall be authenticated by a responsible officer of the contractor. In addition to recording test requirements, methods of verification, results obtained and equipment used, the test report shall contain the following:

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a. Manufacturer's top (assembly) drawing.
b. Assembly, installation and operation instructions.
c. Eight by ten inch color photographs of the grit blast machine.
d. Contractor's quality conformance inspection procedures and documentation.

4.2.3 Examination. Examination of the grit blast machine shall be conducted to conform adherence to the requirements as specified 3.2 through 3.13. This examination shall consist of visual inspection of the grit blast machine and review of the design and manufacturing drawings. A functional check shall be made of the grit blast machine to insure that it is operational. The examinations shall be accomplished using a contractor developed checklist that shall identify all specific grit blast machine attributes and characteristics for which inspection is made, how conformance is determined and results obtained.

4.2.4 Testing. The grit blast machine shall be tested to confirm adherence to the requirements as specified in 3.2 and 3.9. Test method, precision and instrumentation shall be of engineering quality. These tests shall be accomplished using contractor developed procedures and results shall be reported to the government.

4.3 Conformance inspection. Unless otherwise specified (see 6.2), conformance inspection shall be applied to production items offered for acceptance under the contract and shall consist of examination of the units to ensure compliance with the requirements of paragraph 3.1. Failure of any unit to pass any examination or test shall be cause for rejection.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The grit blast machine compressor housing assembly is intended for use on aircraft parts, aircraft sections and ground support equipment.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. First article, if required (see 3.3).
d. Paint requirements if different (see 3.5).
e. Electrical requirements, if different (see 3.11)
f. Initial voltage, if different (see 3.11).
g. Nameplate, if different (see 3.13)
h. Lubrication plate, if required (see 3.14).
i. First article inspection, if different (see 4.2).
j. Conformance inspection, if different (see 4.3).
k. Packaging requirements, if different (see 5.1).

6.3 Subject term (key work) listing.

Grit blast
Dust collector
Exhaust fan
Physical characteristics
Rubber
Corrosion
Storage
Environment

Custodians:
Air Force - 99

Preparing activity:
Air Force - 82

Agent:
Air Force - 99

(Project 4940-0737)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-32031	2. DOCUMENT DATE *(YYMMDD)* 980708

3. DOCUMENT TITLE

GRIT-BLAST MACHINE FOR COMPRESSOR HOUSING ASSEMBLY (T56 Series III)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME SA-ALC/TILDD	b. TELEPHONE *(Include Area Code)* (1) Commercial (210) 925-6314 (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 485 Quentin Roosevelt Rd. Bldg 171 Kelly AFB TX 78241-6524	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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